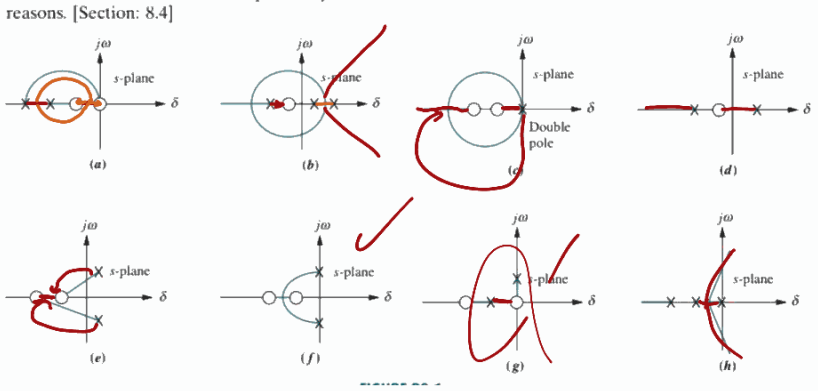
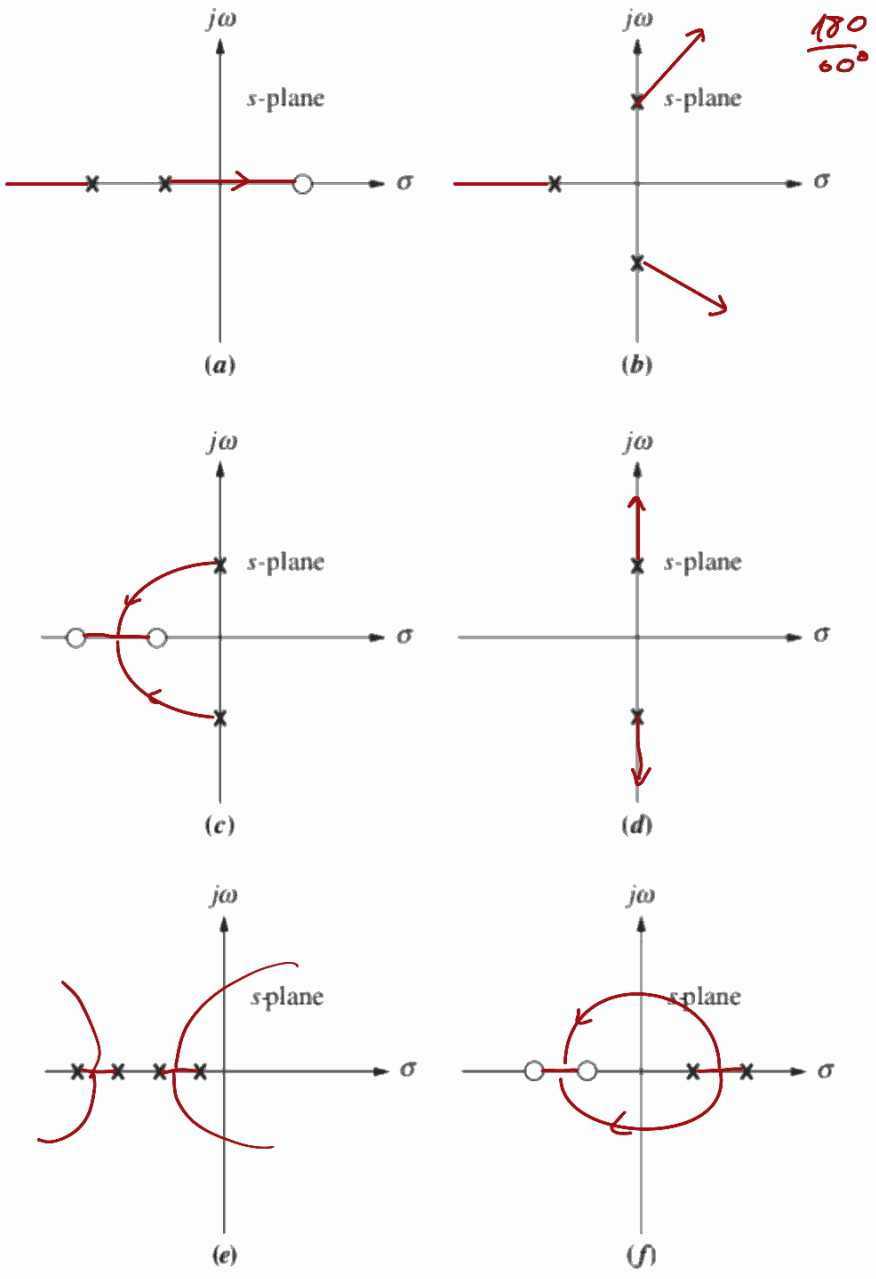
CH8

1. For each of the root loci shown in Figure P8.1, tell whether or not the sketch can be a root locus. If the sketch cannot be a root locus, explain why. Give *all* reasons. [Section: 8.4]



2. Sketch the general shape of the root locus for each of the open-loop pole-zero plots shown in Figure P8.2. [Section: 8.4]

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3. Sketch the root locus for the unity feedback system shown in Figure P8.3 for the following transfer functions: [Section: 8.4]

a. $G(s) = \dfrac{K(s+2)(s+6)}{s^2+8s+25}$

b. $G(s) = \dfrac{K(s^2+4)}{(s^2+1)}$

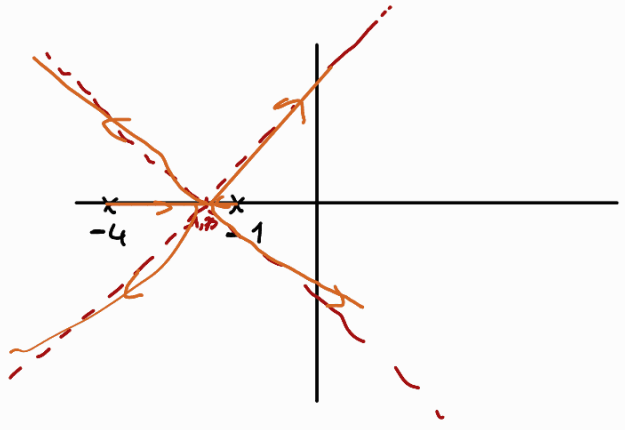
c. $G(s) = \dfrac{K(s^2+1)}{s^2}$

d. $G(s) = \dfrac{K}{(s+1)^3(s+4)}$

$s = -1$ 3 pole

$s = -4$

$\dfrac{\Sigma p - \Sigma z}{n-m} = -\dfrac{7}{4} = -1{,}75$



6. For the open-loop pole-zero plot shown in Figure P8.4, sketch the root locus and find the break-in point. [Section: 8.5]

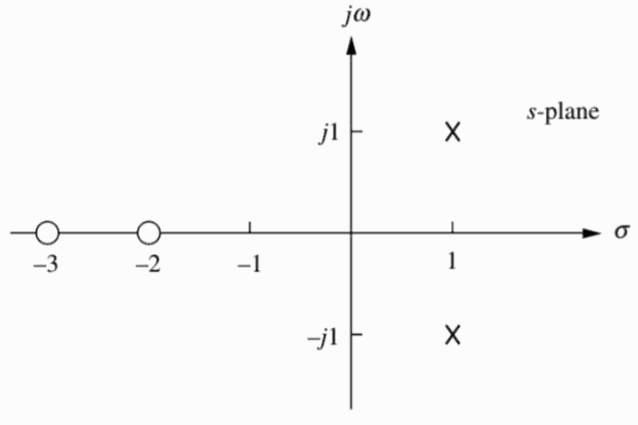


FIGURE P8.4

9. Figure P8.5 shows open-loop poles and zeros. There are two possibilities for the sketch of the root locus. Sketch each of the two possibilities. Be aware that only one can be the *real* locus for specific open-loop pole and zero values. [Section: 8.4]

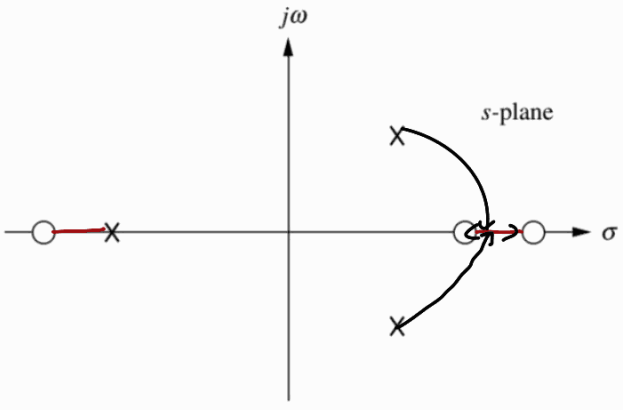


FIGURE P8.5

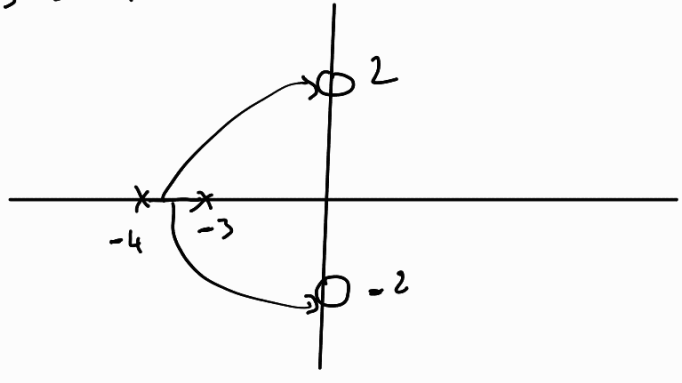
12. Sketch the root locus for the unity feedback system shown in Figure P8.3, where

$$G(s) = \frac{K(s^2+2)}{(s+3)(s+4)}$$

Give the values for all critical points of interest. Is the system ever unstable? If so, for what range of K? [Section: 8.5]

$s = \pm 2j$

$s = -3 \quad s = -4$



23. For the system of Figure P8.8(a), sketch the root locus and find the following: [Section: 8.7]

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a. Asymptotes

b. Breakaway points

c. The range of K for stability

d. The value of K to yield a 0.7 damping ratio for the dominant second-order pair

To improve stability, we desire the root locus to cross the $j\omega$-axis at $j5.5$. To accomplish this, the open-loop function is cascaded with a zero, as shown in Figure P8.8(b).

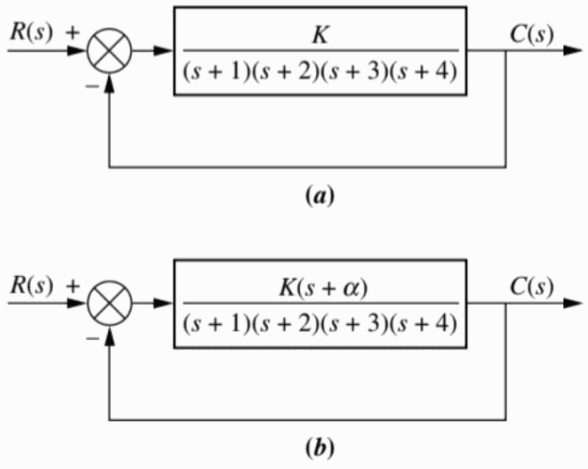


FIGURE P8.8

e. Find the value of α and sketch the new root locus.

f. Repeat Part **c** for the new locus.

g. Compare the results of Part **c** and Part **f**. What improvement in transient response do you notice?

27. For the unity feedback system shown in Figure P8.3, where

$$G(s) = \frac{K(s^2 - 2s + 2)}{(s+2)(s+4)(s+5)(s+6)}$$

do the following: [Section: 8.7]

a. Sketch the root locus.

b. Find the asymptotes.

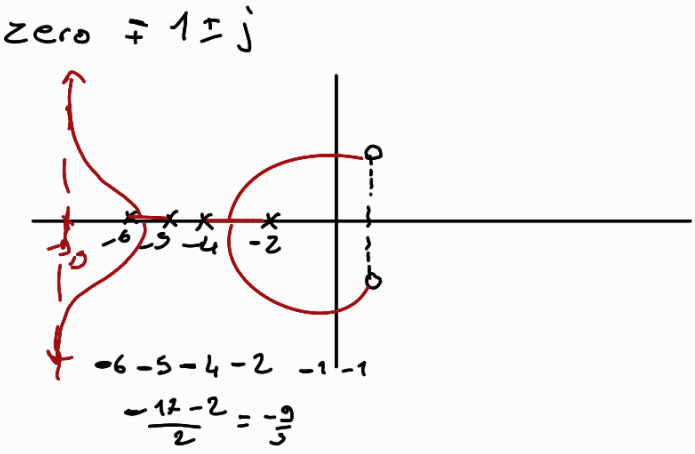
c. Find the range of gain, K, that makes the system stable.

d. Find the breakaway points.

e. Find the value of K that yields a closed-loop step response with 25% overshoot.

f. Find the location of higher-order closed-loop poles when the system is operating with 25% overshoot.

g. Discuss the validity of your second-order approximation.



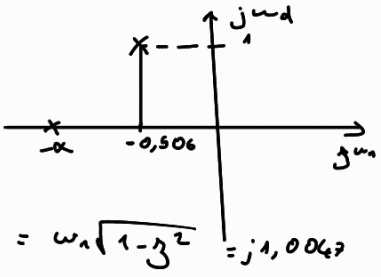
31. For the unity feedback system shown in Figure 8.3, where

$$G(s) = \frac{K(s+6)}{(s^2 + 10s + 26)(s+1)^2(s+\alpha)}$$

design K and α so that the dominant complex poles of the closed-loop function have a damping ratio of 0.45 and a natural frequency of 9/8 rad/s.

$\zeta = 0,45 \quad \omega_n = \frac{9}{8}$

$\zeta\omega_n = 0,506$



$\omega_d = \omega_n\sqrt{1-\zeta^2} = j\,1,0047$

Sayfa 535

5)

5. Consider the unity feedback system shown in Figure P9.1 with

$$G(s) = \frac{K}{(s+3)(s+5)(s+7)}$$

a. Design a compensator that will yield $K_p = 20$ without appreciably changing the dominant pole location that yields a 10% overshoot for the uncompensated system. [Section: 9.2]

$M_p = 0,1 = e^{-\frac{\pi}{\omega_d}\sigma \to \zeta\omega_n}$

$0,1 = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$

$\ln 0,1 = \frac{-\pi\zeta}{\sqrt{1-\zeta^2}}$

$0,73 = \frac{\zeta}{\sqrt{1-\zeta^2}}$

$0,537 = \frac{\zeta^2}{1-\zeta^2}$

$0,537 - 0,537\zeta^2 = \zeta^2$

$\zeta = 0,59$

8. Design a PD controller for the system shown in Figure P9.2 to reduce the settling time by a factor of 4 while continuing to operate the system with 20% overshoot. Compare your performance to that obtained in Example 9.7.

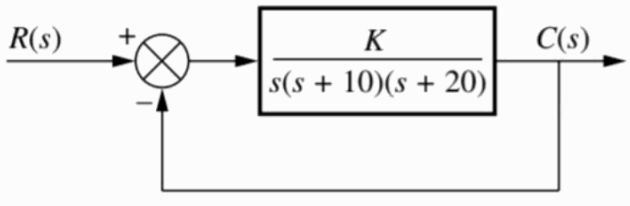


FIGURE P9.2

$M_p = 0,2 = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$

$\Rightarrow \zeta = 0,46$

$t_s = \frac{-\ln x}{\zeta\omega_n} \to \times 4$

Example 9.1

Effect of an Ideal Integral Compensator

PROBLEM: Given the system of Figure 9.4(*a*), operating with a damping ratio of 0.174, show that the addition of the ideal integral compensator shown in Figure 9.4(*b*) reduces the steady-state error to zero for a step input without appreciably affecting transient response. The compensating network is chosen with a pole at the origin to increase the system type and a zero at −0.1, close to the compensator pole, so that the angular contribution of the compensator evaluated at the original, dominant, second-order poles is approximately zero. Thus, the original, dominant, second-order closed-loop poles are still approximately on the new root locus.

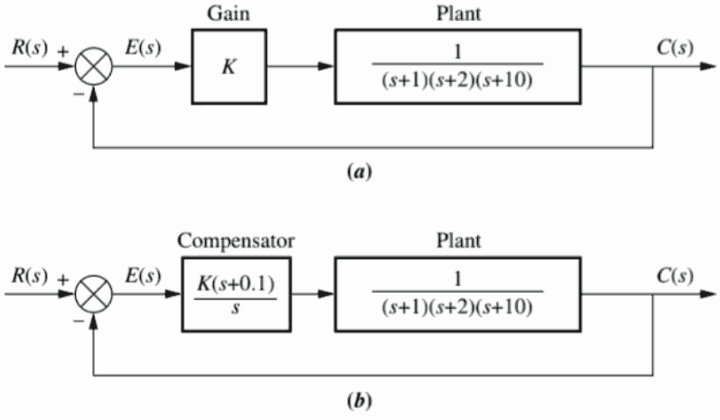


FIGURE 9.4 Closed-loop system for Example 9.1: **a.** before compensation; **b.** after ideal integral compensation

$k = 164,6$

$kp = \lim_{s \to 0} G(s) = \frac{164,6}{20} = 8,23$

$e(\infty) = \frac{1}{1+kp} = 0,108$

23. For the system of Figure P8.8(*a*), sketch the root locus and find the following: [Section: 8.7]

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a. Asymptotes

b. Breakaway points

c. The range of K for stability

d. The value of K to yield a 0.7 damping ratio for the dominant second-order pair

To improve stability, we desire the root locus to cross the $j\omega$-axis at $j5.5$. To accomplish this, the open-loop function is cascaded with a zero, as shown in Figure P8.8(*b*).

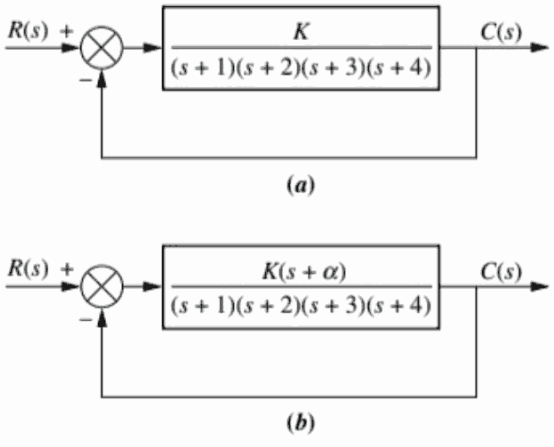


FIGURE P8.8

e. Find the value of α and sketch the new root locus.

f. Repeat Part **c** for the new locus.

g. Compare the results of Part **c** and Part **f**. What improvement in transient response do you notice?

centroid: Σ poles − Σ zeros

$= -1-2-3-4$

$= -10$

$\alpha_1 = \frac{(2k+1)\pi}{4}$

b) $1 + \frac{K}{(s+1)(s+2)(s+3)(s+4)}$

$K + (s+1)(s+2)(s+3)(s+4) = 0$

$$k = -s^4 + 10s^3 + 35s^2 + 50s + (K+24)$$

$$\frac{dk}{ds} = -4s^3$$